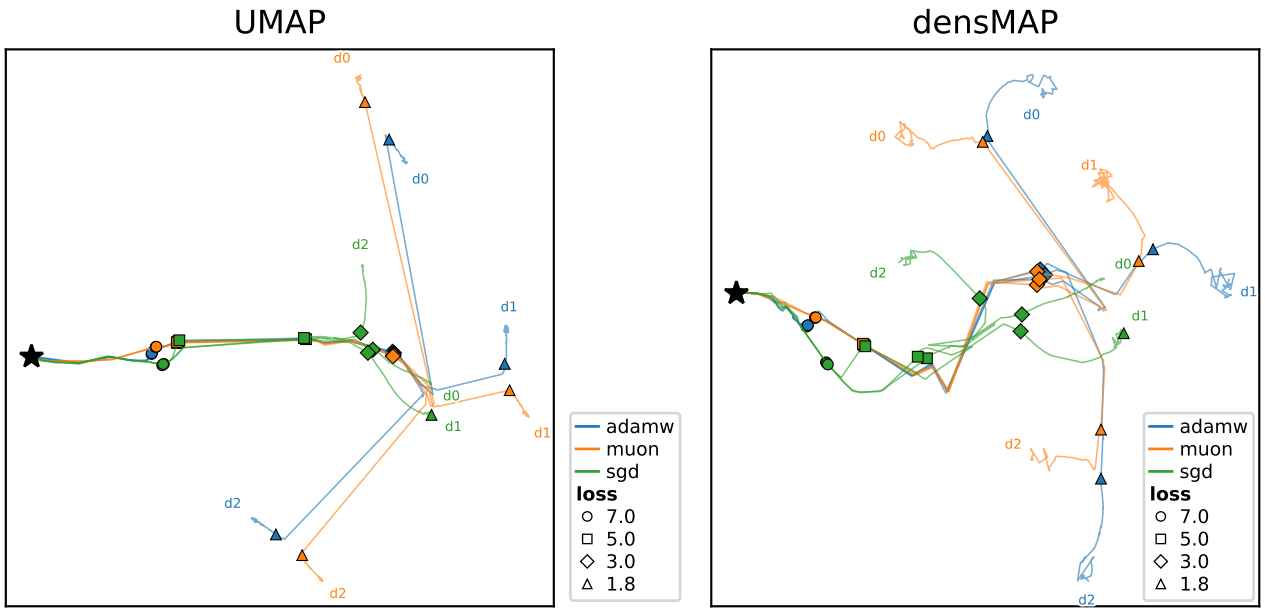
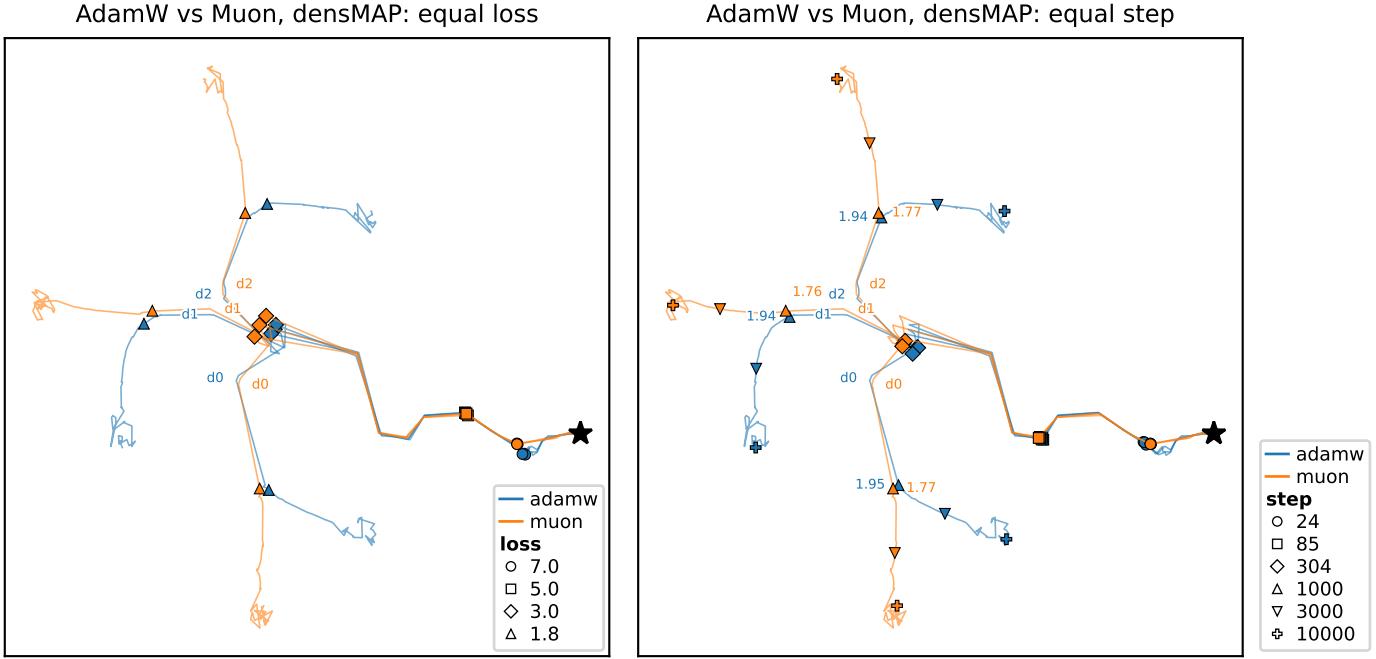
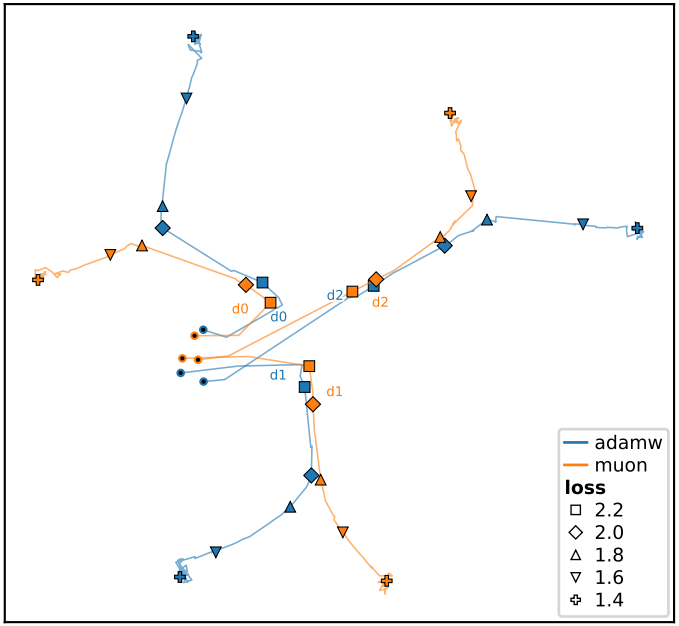
Setting (all pages): TinyStories-10M. nanoGPT, 6 layers, 6 heads, width 384, 13.9M parameters; BPE vocabulary 4096, context 256 tokens. Batch 128 sequences × 256 tokens; 10,000 steps = 328M tokens (≈0.6 epochs). Linear warmup over 200 steps, then cosine decay to 10% of the peak rate. Learning rates: SGD 0.03 (momentum 0.9), AdamW 2×10^{-3}, Muon 0.01. Shared initialization; three data orders d0–d2 per optimizer (of the five trained). Distance between two (run, step) points: square root of the Jensen–Shannon divergence of their next-token distributions on 512 fixed validation positions. Training loss on 64 fixed training sequences.

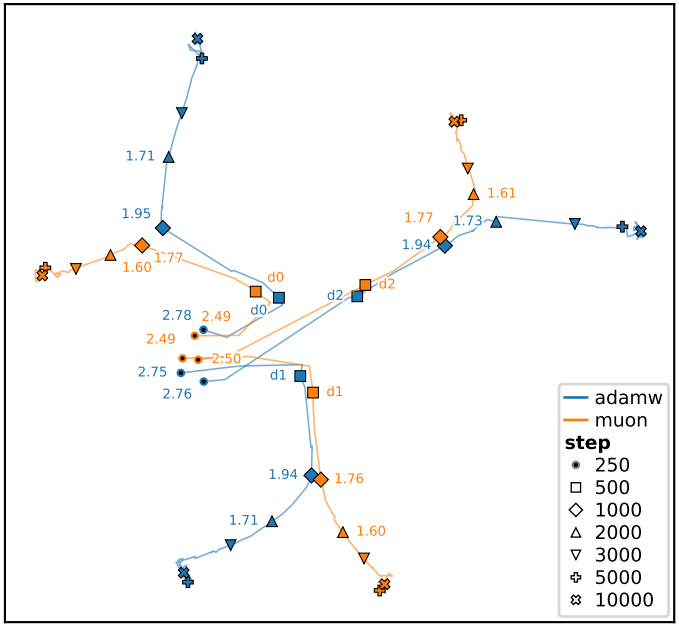


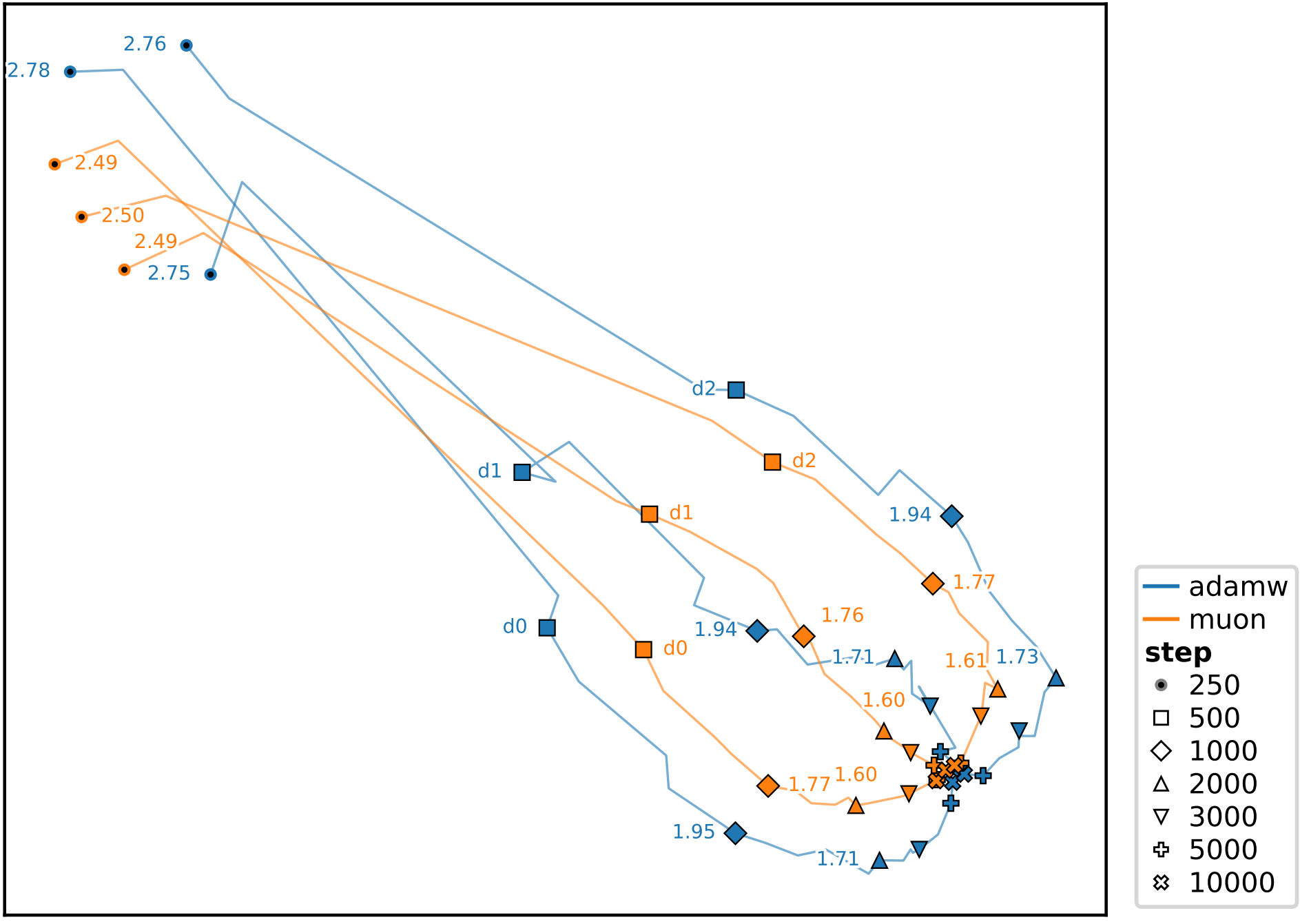

AdamW vs Muon, densMAP: equal loss
AdamW vs Muon, densMAP: equal step
d2
d1
d0
1.94
1.77
1.76
1.95
adamw
muon
loss
7.0
5.0
3.0
1.8
step
24
85
304
1000
3000
10000

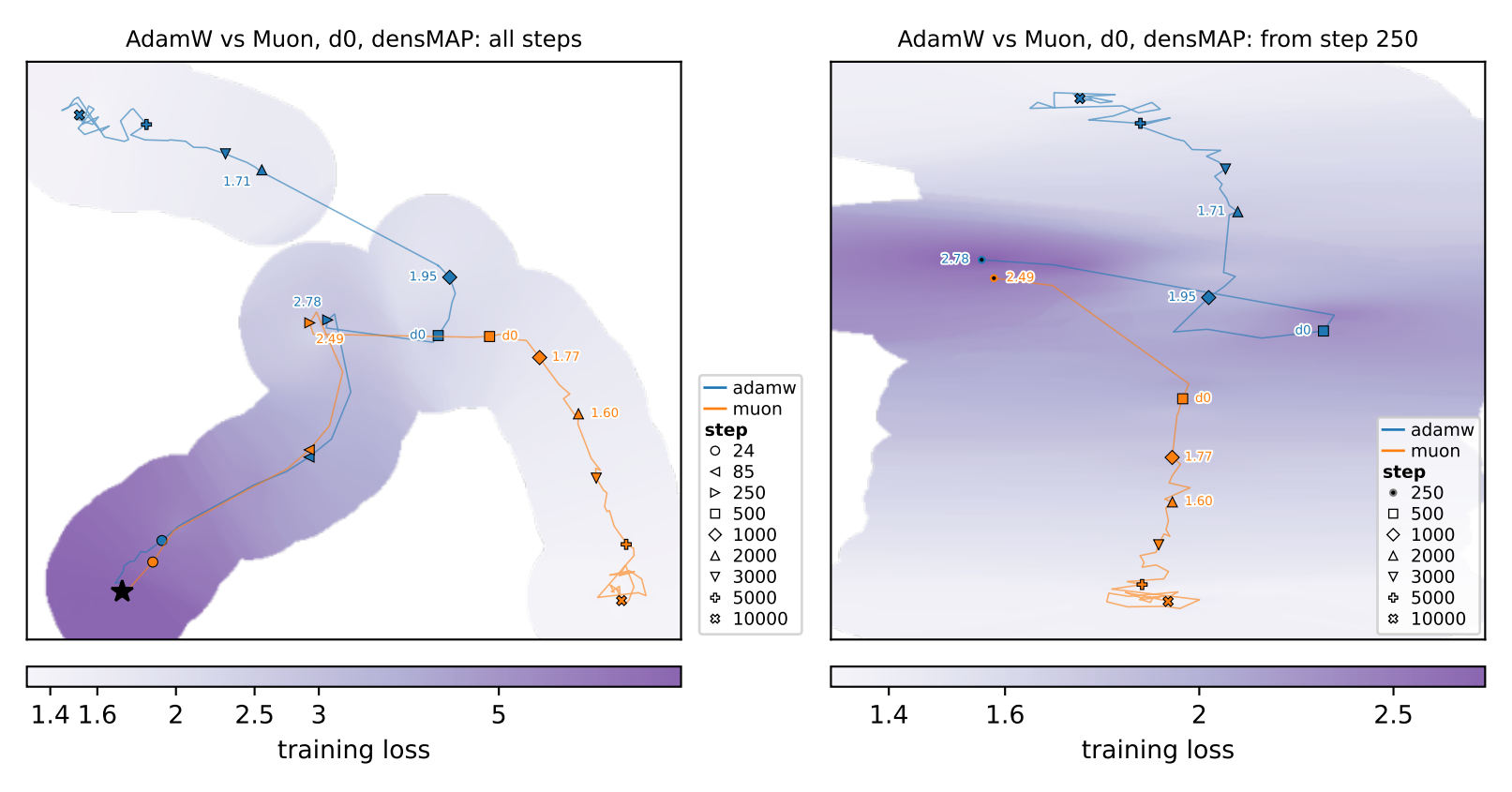
AdamW vs Muon, densMAP: equal loss, from step 250

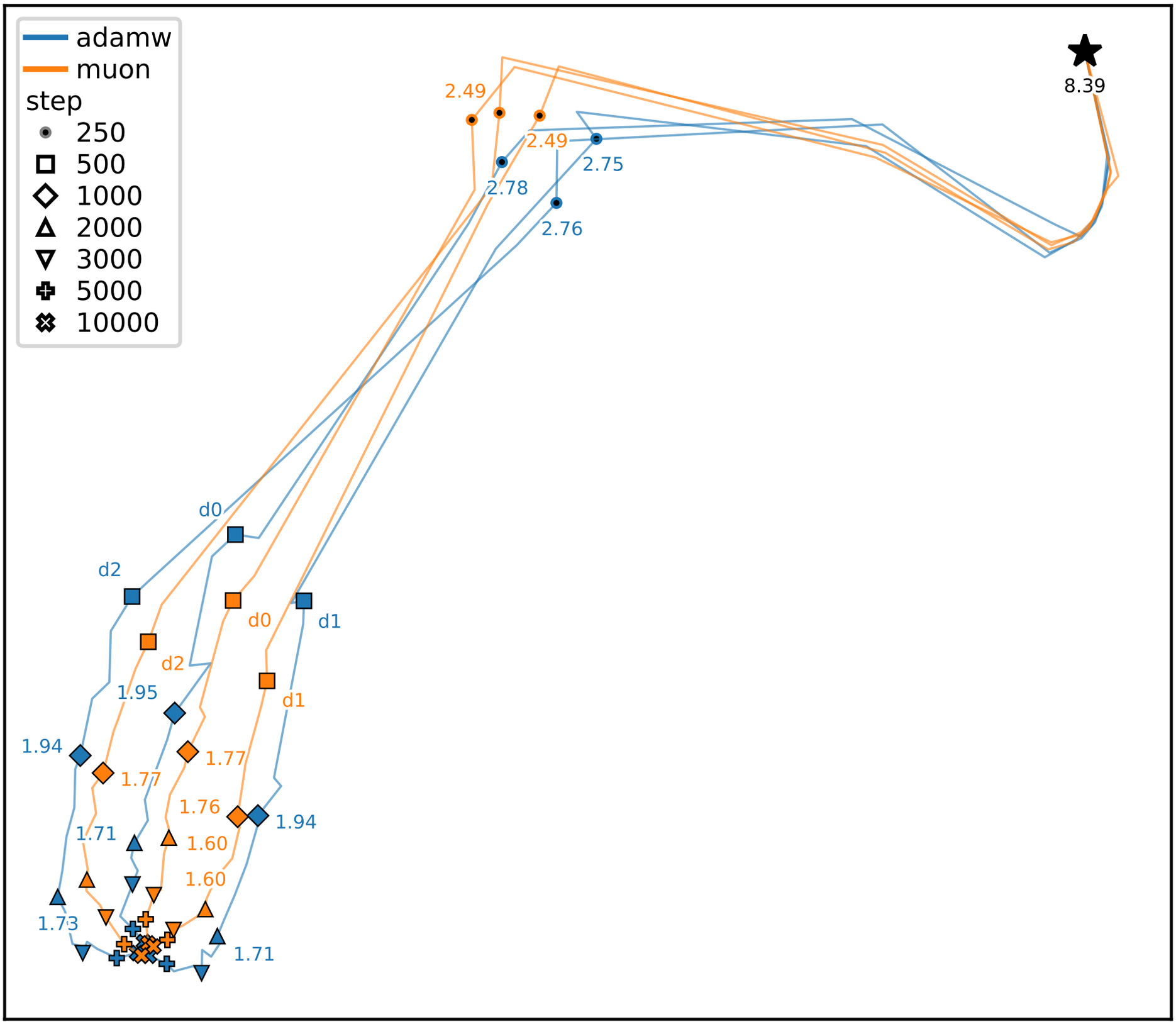


AdamW vs Muon, densMAP: equal step, from step 250




AdamW vs Muon, densMAP: equal step, from step 250
(thinned, δ = 0.28)
2.76
2.78
2.49
2.50
2.49
2.75
d2
d2
d1
d1
1.94
1.77
d0
1.94
1.76
d0
1.71
1.61
1.73
1.60
1.77
1.60
1.95
1.71
adamw
muon
step
250
500
1000
2000
3000
5000
10000


AdamW vs Muon, d0, densMAP: all steps
AdamW vs Muon, d0, densMAP: from step 250
1.71
1.95
2.78
2.49
d0
1.77
1.60
adamw
muon
step
24
85
250
500
1000
2000
3000
5000
10000
1.4
1.6
2
2.5
3
5
training loss


AdamW and Muon, thinned densMAP from step 0 (star: shared initialization)
adamw
muon
step
250
500
1000
2000
3000
5000
10000
8.39
2.49
2.49
2.75
2.78
2.76
d0
d2
d0
d1
d2
d1
1.95
1.94
1.77
1.77
1.76
1.94
1.71
1.60
1.60
1.73
1.71